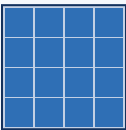


Three forms, one factorization

All three reconstruct A exactly. They differ in how much of U and Σ is kept — and what is dropped is precisely the part that multiplies against zero singular values.

EVERYTHING KEPT

1



Full SVD | U and V both square and orthogonal

SHAPES	WHAT IS KEPT	WHAT IS DROPPED	USED FOR
$U : m \times m, \Sigma : m \times n, V : n \times n$	all of U, V , and a rectangular Σ	nothing	theory, the four subspaces

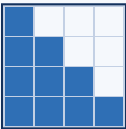
The only form where U and V are genuinely orthogonal matrices, so $UU^T = I$ as well as $U^T U = I$. That is what makes it the form to state results with — the last $m - r$ columns of U span the left null space and would otherwise be gone.

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TRIMMED TO WHAT MULTIPLIES

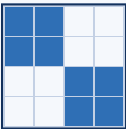
2



Thin SVD | $m > n$ — keep n columns of U

SHAPES	WHAT IS KEPT	WHAT IS DROPPED	USED FOR
$U : m \times n, \Sigma : n \times n, V : n \times n$	the first n columns of U	the last $m - n$ columns of U	least squares on tall matrices

The dropped columns multiply rows of Σ that are entirely zero, so removing them changes nothing about the product. Note U is no longer square: $U U^T = I$ still holds, but $U^T U$ does not.



Compact SVD | keep only the r nonzero singular values

SHAPES	WHAT IS KEPT	WHAT IS DROPPED
$U : m \times r, \Sigma : r \times r, V : n \times r$	the rank- r content only	every zero singular value and its vectors

USED FOR $r \times r$
storage, rank-deficient matrices

The most economical exact form: Σ is square with a strictly positive diagonal, so it is invertible — which is what makes the **pseudoinverse** computable as $V \Sigma^{-1} U^T$.

Nothing is approximated anywhere in this table. The full form carries orthonormal bases for all four **fundamental subspaces**, which is why it is the one to state theorems with; the compact form carries only what contributes to A , which is why it is the one to store. Approximation begins only when the compact form is truncated below rank r .